Day 1: Environment Setup & Lab Foundations

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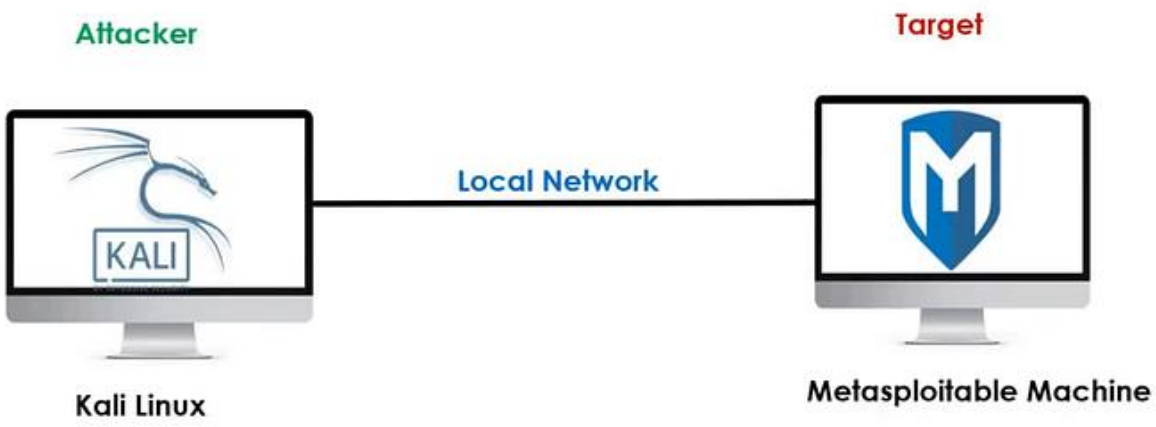
1. Introduction

To kick off the **YoungDevInterns_Ethical_Hacking_Internship**, the first day is all about building a safe, legal, and powerful environment. In the world of cybersecurity, you cannot practice hacking on real-world websites or live networks without explicit permission. Therefore, we build a "sandbox" an isolated digital playground where we can learn and fail safely.

2. Objective of the Day

The primary goal is to transform a standard computer into a professional penetration testing station. This involves setting up a **Hypervisor** to host multiple operating systems and creating a private network so the "Attacker" machine can interact with the "Victim" machine without any risk to the outside world.



3. Tools & Software Used

- **VirtualBox:** A powerful, free virtualization tool that allows us to run multiple "Virtual Machines" (VMs) simultaneously.
- **Kali Linux:** The industry-standard operating system for hackers, pre-loaded with over 600 security tools.
- **Metasploitable2:** An intentionally vulnerable Linux-based virtual machine used specifically for security testing.

4. Step-by-Step Implementation

a) Virtualization Setup

First, I installed **Oracle VM VirtualBox**. This software acts as the hardware layer for our virtual lab. It allows us to allocate specific amounts of RAM, CPU, and Disk space to our guest operating systems.

b) Installing the Attacker (Kali Linux)

I downloaded the pre-configured Virtual Machine image from the official Kali website.

- I imported the .ova file into VirtualBox.
- I adjusted the settings to provide **2GB of RAM** and **2 CPU cores** to ensure the OS runs smoothly during heavy scanning tasks.

c) Installing the Victim (Metasploitable2)

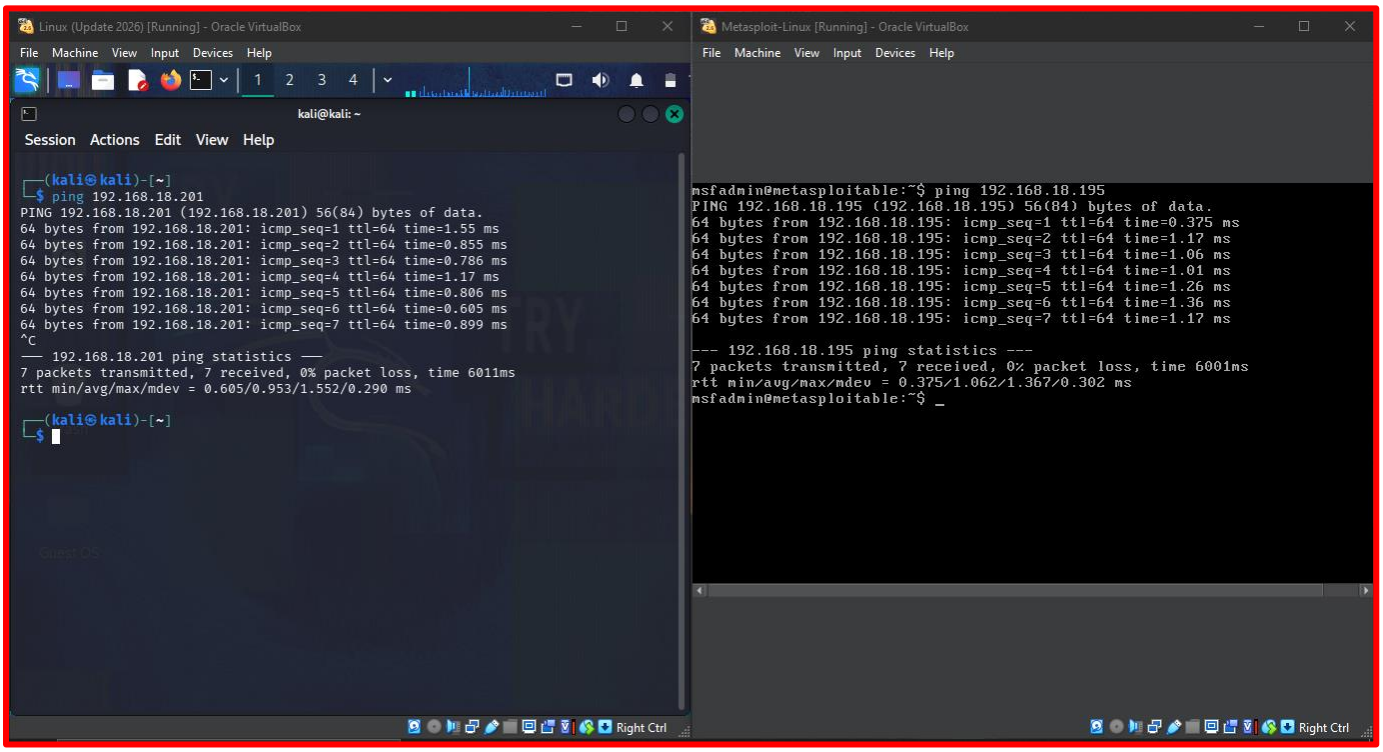
Since Metasploitable2 is a target machine, I set it up using its virtual disk file (.vmdk).

- I created a new "Linux" machine in VirtualBox.
- I selected "Use an existing virtual hard disk file" and pointed it to the Metasploitable file.

5. Network Configuration & Security

This is the most critical step for any ethical hacker.

- I configured both machines to use a **Host-Only Network** (or a private NAT Network).
- **Reason:** This ensures the two machines can "talk" to each other, but my hacking activity stays completely isolated from my local home Wi-Fi and the public internet. This prevents accidental damage or legal issues.



6. What I Did Today

1. **Environment Build:** Successfully installed the Hypervisor (VirtualBox).
2. **OS Deployment:** Deployed Kali Linux as my primary hacking platform.
3. **Target Setup:** Set up Metasploitable2 to serve as my practice target.
4. **Connectivity Check:** Configured the virtual network adapters so both machines reside on the same internal subnet.

7. What I Learned

- **The Concept of Virtualization:** I learned how to run an entire OS as a "guest" without affecting my "host" (main) computer.
- **The Attacker vs. Victim Lab Model:** In ethical hacking, you always need a target. Metasploitable serves as a safe environment to test exploits without breaking the law.
- **Network Isolation:** I learned the importance of **Isolating the Lab**. Setting up a private network prevents accidental "scanning" of my home router, which is a crucial professional habit.
- **Snapshots:** I learned how to take a "Snapshot" in VirtualBox. If I accidentally delete a critical file while hacking, I can "teleport" the machine back to its original working state with one click.

Key Takeaway for Newbies: *"A good hacker is only as good as their lab. Setting this up correctly is the most important step to prevent legal issues and system crashes."*

8. Hands-on Examples & Verification

- **Command used to check IP:** ifconfig
  - *Usage:* I ran this on both machines to ensure they are on the same subnet (e.g., both starting with 192.168.56.x).
- **Verification:** I successfully "pinged" the Metasploitable machine from my Kali terminal to confirm they are connected.